HUFFMAN ENCODING USING GREEDY METHOD

Theory • Algorithm • Example • Complexity • Viva

1. Theory

Huffman Encoding is a **lossless data compression technique** used to reduce data size. It assigns variable-length binary codes to characters based on their frequency. Frequently occurring characters receive shorter codes, while less frequent characters receive longer codes.

It uses the **Greedy Algorithm** to construct a Huffman Tree. The algorithm repeatedly selects the two nodes with the lowest frequencies and combines them into a new parent node until only one node remains. A **Min Heap (Priority Queue)** efficiently selects the two smallest-frequency nodes.

2. Algorithm Steps

1. Accept characters and their frequencies.

2. Create a node for each character.

3. Insert all nodes into a Min Heap.

4. Remove the two nodes with the smallest frequencies.

5. Create a parent node with their combined frequency.

6. Assign the removed nodes as left and right children.

7. Insert the new node back into the Min Heap.

8. Repeat until only one node remains.

9. Traverse the Huffman Tree to generate binary codes.

10. Assign 0 to the left branch and 1 to the right branch.

3. Example

Consider these characters and frequencies:

Character	Frequency
A	5
B	9
C	12
D	13
E	16

Combining the two smallest frequencies:
A(5) + B(9) = 14
C(12) + D(13) = 25
14 + E(16) = 30
25 + 30 = 55

One possible set of Huffman codes:

Character	Huffman Code
A	100
B	101
C	00
D	01

Character	Huffman Code
E	11

Note: Codes can vary with node arrangement while remaining optimal.

4. Complexity Analysis

Time Complexity: O(n log n), because heap operations take O(log n) time.
Space Complexity: O(n), for the Huffman Tree and Min Heap.

5. Applications

• File compression.
• Image compression.
• Text compression.
• Data transmission.
• Lossless data storage.

6. Main Theory to Remember

Huffman Encoding is a lossless compression technique that uses the **Greedy Algorithm and Min Heap** to generate optimal prefix-free binary codes. It repeatedly combines the two nodes with the smallest frequencies. Left branch = 0 and right branch = 1.

7. Viva Questions

Q1. What is Huffman Encoding?
A lossless data compression technique that assigns variable-length binary codes.

Q2. Which algorithm is used?
Greedy Algorithm.

Q3. Which data structure is used?
Min Heap (Priority Queue).

Q4. What is a Huffman Tree?
A binary tree used to generate Huffman codes.

Q5. What is prefix-free coding?
No character's code is the prefix of another character's code.

Q6. What is the time complexity?
O(n log n).

8. Conclusion

Huffman Encoding compresses data by assigning shorter codes to frequently occurring characters and longer codes to less frequent characters. It uses a Greedy Algorithm to generate optimal prefix-free codes without losing information.

EXAM-READY LINE

Huffman Encoding uses a Greedy Algorithm and Min Heap to repeatedly combine the two least frequent nodes and generate optimal prefix-free binary codes.